

Contoh Soal:

PERPANGKATAN

```
1. public int power(int base, int exponent) {
2.     if (exponent == 0) {
3.         return 1;
4.     }
5.     return base * power(base,exponent-1);
6. }
```

FAKTORIAL

```
1. public int factorial(int num) {
2.     if (num<=1) {
3.         return 1;
4.     }
5.     return num * factorial(num-1);
6. }
```

PERKALIAN ARRAY

```
1. public int productofArray(int A[], int N)
2.     {
3.         if (N <= 0)
4.             return 1;
5.         return (productofArray(A, N - 1) * A[N - 1]);
6.     }
```

JUMLAH RENTANG NILAI TERTENTU

Misalnya input angkanya 6 berarti jumlahnya = $6 + 5 + 4 + 3 + 2 + 1 = 21$

Misalnya input angkanya 3 berarti jumlahnya = $3 + 2 + 1 = 6$

```
1. public int recursiveRange(int num) {
2.     if (num<=0) {
3.         return 0;
4.     }
5.     return num + recursiveRange(num - 1);
6. }
```

DERET FIBONACCI

```
1. public int fib(int n) {
2.     if (n<0) {
3.         return -1;
4.     }
5.     if (n==0 || n==1) {
6.         return n;
7.     }
8.     return fib(n-1) + fib(n-2);
9. }
```

REVERSE STRING

```
1. public String reverse(String str)
2.     {
3.         if (str.isEmpty())
4.             return str;
5.         //Calling Function Recursively
6.         return reverse(str.substring(1)) + str.charAt(0);
7.     }
```

PALINDROM

```
1. public boolean isPalindrome(String s)
2.     { // if length is 0 or 1 then String is palindrome
3.         if(s.length() == 0 || s.length() == 1)
4.             return true;
5.         if(s.charAt(0) == s.charAt(s.length()-1))
6.             return isPalindrome(s.substring(1, s.length()-1));
7.         return false;
8.     }
```

KARAKTER PERTAMA UPPERCASE

```
1. public static char first(String str) {
2.     if(str.isEmpty()) {
3.         return ' ';
4.     }
5.     if (Character.isUpperCase(str.charAt(0))) {
6.         return str.charAt(0);
7.     }else {
8.         return first(str.substring(1, str.length()));
9.     }
10. }
```

KATA HURUF KAPITAL

```
1. public static String capitalizeWord(String str) {
2.
3.     if(str.isEmpty()) {
4.         return str;
5.     }
6.     char chr = str.charAt(str.length()-1);
7.     if(str.length()==1) {
8.         return Character.toString(Character.toUpperCase(chr));
9.     }
10.    if((str.substring(str.length()-2, str.length()-1).equals(" "))) {
11.        chr = Character.toUpperCase(chr);
12.    }
13.    return capitalizeWord(str.substring(0, str.length()-1))+
        Character.toString(chr);
14. }
```